# Superquadric view factors via Fox $H$ functions

Hao Wu[a,*]

[a] *School of Nuclear Science and Engineering, North China Electric Power University, Beijing 102206, China*

**Abstract**

An analytical geometric-moment formulation is developed for radiative view factors between identical, parallel, side-by-side tangent superquadric particles. The particle geometry follows the $n_1, n_2$ representation used in discrete element simulations, with the complete analytical development established for its axisymmetric subclass. A projected-overlap formulation separates particle geometry from the directional radiation kernel. Explicit Gamma-function moments generate an infinite series whose convergence is governed by the projected difference domain. Direct summation produces a two-dimensional integral with an inverse-trigonometric kernel, while a multivariate Fox–Wright representation leads to Fox $H$ continuation beyond the original convergence domain. Conjugate boundary values recover the geometrically admissible angular support. The spherical limit additionally gives explicit $n^{-3}$ term decay and $N^{-2}$ truncation-error laws. Numerical checks verify the moment coefficients, summed representation and continued Barnes integral independently. Diffuse-ray comparisons recover exact spheroidal exchange and quantify the projection approximation for other shapes. For the sampled equal-axis particles with $1 \leq m \leq 4$, the relative discrepancy remains below approximately 4.5%, while larger deviations toward the cylindrical limit identify the limits of the geometric closure.

*Keywords:* Superquadric particles, Radiative view factor, Geometric moments, Series summation, Fox–Wright function, Fox $H$ function

## 1. Introduction

Analytical view factors provide reference solutions for radiative heat-transfer calculations and reveal how geometric parameters influence energy exchange. Classical contour formulations reduce surface integrals to boundary integrals [1]. Subsequent studies developed accurate planar-surface calculations [2], general procedures for arbitrary planar polygons [3], analytical polygon-to-polygon expressions [4], and projected-solid-angle formulas based on the Gauss–Bonnet theorem [5]. These contributions demonstrate the value of transforming the original radiation integral before numerical evaluation.

Analytical treatments of curved surfaces include sphere-to-axisymmetric-body relations [6, 7], configuration factors for curved enclosures [8], spherical-cap solutions [9], and spheroid and ellipse view factors from a plate element [10]. These geometries require explicit treatment of projection and visibility. In particular, the recent spheroidal formulation shows how a curved-body problem can be connected to an equivalent projected ellipse. For particle pairs, the approximate sphere-to-sphere relation of Felske [11] and the exact spherical solution of Degweker [12] provide useful benchmarks.

Superquadrics extend these standard geometries through a small set of shape parameters. Their implementation in discrete element simulations [13] motivates analytical radiation models using the same geometric description. Such models complement ray tracing, whose efficiency and accuracy depend on geometry, visibility and the magnitude of the view factor [14, 15].

*Corresponding author.
*Email address:* wuhao@ncepu.edu.cn (Hao Wu)

The present work establishes three connected representations for tangent axisymmetric superquadric particles: a geometric-moment series, its directly summed transcendental integral, and a multivariate Fox $H$ continuation. Fox $H$ functions and Barnes integrals provide the mathematical framework for representing Gamma-function series beyond their original convergence domains [16–19]. The specific contribution here is to connect that framework to projected particle geometry and to determine the real continuation through the admissible radiation directions.

The analysis distinguishes the exact evaluation of a stated geometric model from the approximation involved in applying that model to a non-ellipsoidal particle. This distinction is examined through independent numerical checks and diffuse-ray calculations.

## 2. Theoretical model

### *2.1. Superquadric particles*

The particle surface follows the LIGGGHTS convention [13]:

$$\left(\left|\frac{x}{a}\right|^{n_2} + \left|\frac{y}{b}\right|^{n_2}\right)^{n_1/n_2} + \left|\frac{z}{c}\right|^{n_1} = 1, \tag{1}$$

where $a, b, c$ are the principal semi-axes and $n_1, n_2$ control particle shape. The choice $n_1 = n_2 = 2$ gives an ellipsoid.

The complete analytical development below concerns

$$a = b, \qquad n_2 = 2, \qquad n_1 = m,$$

so that

$$\boxed{\left(\frac{\rho}{a}\right)^m + \left|\frac{z}{c}\right|^m = 1, \qquad \rho = \sqrt{x^2 + y^2}, \qquad \lambda = \frac{c}{a}.} \tag{2}$$

The convex mathematical family has $m \geq 1$. Two identical particles have parallel symmetry axes along $z$, with their centers separated by $h = 2a$ along $x$. They therefore touch in a side-by-side arrangement.

This parameter path approaches a circular cylinder as $m \to \infty$. Increasing both $n_1$ and $n_2$ instead produces a cuboid-like limit.

The projected area normal to the center-to-center direction and the total surface area are

$$A_p = 4a^2 \lambda G_m, \qquad A_s = 4\pi a^2 \lambda I_m(\lambda), \tag{3}$$

where

$$G_m = \frac{\Gamma^2(1 + 1/m)}{\Gamma(1 + 2/m)}$$

and

$$I_m(\lambda) = \int_0^1 (1 - t^m)^{1/m} \sqrt{1 + \lambda^{-2} t^{2m-2} (1 - t^m)^{2/m-2}}\, dt. \tag{4}$$

Both geometric quantities retain the exact particle shape.

*2.2. View-factor formulation and projected-overlap closure*

For diffuse radiation, the exact view factor is

$$F_{12} = \frac{1}{\pi A_s} \int_{A_1} \int_{A_2} \chi_{12} \frac{[\mathbf{n}_1 \cdot \widehat{\mathbf{R}}]_+ [-\mathbf{n}_2 \cdot \widehat{\mathbf{R}}]_+}{R^2} \, dA_2 \, dA_1, \tag{5}$$

where $[x]_+ = \max(x, 0)$, $\mathbf{R}$ joins the surface points, and $\chi_{12}$ accounts for visibility. The particles are isolated from third-body obstruction.

An equivalent directional formulation integrates the overlap of particle projections on planes perpendicular to the radiation direction. Introduce the affine transformation

$$\mathbf{r} = a \, \mathrm{diag}(1, 1, \lambda) \, \bar{\mathbf{r}}, \qquad \beta = 1 - \lambda^2.$$

For a reference direction $\bar{\boldsymbol{\omega}}$, the combined transformation of projected area and solid angle contributes

$$\frac{a^2 \lambda^2}{(1 - \beta \bar{\omega}_z^2)^2}.$$

The exact directional representation becomes

$$F_{12} = \frac{a^2 \lambda^2}{\pi A_s} \int_{\bar{\omega}_x > 0} \frac{\mathcal{O}_m(\bar{\boldsymbol{\omega}})}{(1 - \beta \bar{\omega}_z^2)^2} \, d\bar{\Omega}, \tag{6}$$

where $\mathcal{O}_m$ denotes the overlap area of the projected reference particles, whose centers are separated by two reference length units.

Define the meridional domain

$$D_m = \{(u, v) : |u|^m + |v|^m \leq 1\}, \qquad |D_m| = 4G_m,$$

and its overlap function

$$C_m(s, t) = \int_{\mathbb{R}^2} \mathbf{1}_{D_m}(u, v) \mathbf{1}_{D_m}(u - s, v - t) \, du \, dv. \tag{7}$$

The adopted geometric closure is

$$\mathcal{O}_m(\bar{\boldsymbol{\omega}}) \approx C_m(2\bar{\omega}_y, 2\bar{\omega}_z).$$

It represents directional overlaps using a fixed reference silhouette. For $m = 2$, the reference particle is a sphere and every projection is a unit disk. The closure is then exact, and the affine transformation recovers spheroidal exchange for arbitrary $\lambda$. For $m \neq 2$, directional changes in the projected silhouette are approximated.

Writing

$$\xi = \bar{\omega}_y, \qquad \eta = \bar{\omega}_z,$$

gives

$$\boxed{\widehat{F}_{12} = \frac{\lambda}{4\pi^2 I_m(\lambda)} \iint_{\xi^2 + \eta^2 < 1} \frac{C_m(2\xi, 2\eta)}{\sqrt{1 - \xi^2 - \eta^2}(1 - \beta\eta^2)^2} \, d\xi \, d\eta.} \tag{8}$$

The hat distinguishes the projected-overlap model from the exact view factor for a general superquadric.

*2.3. Geometric moments and infinite series*

Let $(U, V)$ be uniformly distributed in $D_m$. Its even moments follow from Beta integrals:

$$\boxed{\mu_{pq}^{(m)} = \langle U^{2p} V^{2q} \rangle = \frac{\Gamma\left(\frac{2p+1}{m}\right)\Gamma\left(\frac{2q+1}{m}\right)\Gamma\left(1+\frac{2}{m}\right)}{\Gamma^2\left(\frac{1}{m}\right)\Gamma\left(1+\frac{2p+2q+2}{m}\right)}.} \tag{9}$$

For two independent samples, define

$$\xi = \frac{U - U'}{2}, \qquad \eta = \frac{V - V'}{2}.$$

Their difference moments are

$$\boxed{M_{pq}^{(m)} = \frac{1}{4^{p+q}} \sum_{i=0}^{p} \sum_{j=0}^{q} \binom{2p}{2i} \binom{2q}{2j} \mu_{ij}^{(m)} \mu_{p-i,q-j}^{(m)}.} \tag{10}$$

Every geometric coefficient is therefore explicit.

Expanding the directional kernel gives

$$\boxed{F_\Sigma = F_0 \sum_{p,q,r \geq 0} \frac{(1/2)_{p+q}}{p!\, q!} (r+1) \beta^r M_{p,q+r}^{(m)},} \tag{11}$$

where

$$F_0 = \frac{A_p^2}{\pi A_s h^2} = \frac{\lambda G_m^2}{\pi^2 I_m(\lambda)}.$$

Here $(a)_n$ is the Pochhammer symbol. Collecting terms of the same geometric order gives

$$F_\Sigma = F_0 \sum_{n=0}^{\infty} R_{2n},$$

$$\boxed{R_{2n} = \sum_{p+q+r=n} \frac{(1/2)_{p+q}}{p!\, q!} (r+1) \beta^r M_{p,q+r}^{(m)}.} \tag{12}$$

The leading term is $R_0 = 1$. Higher orders account for the finite extent of the projected particles.

*2.4. Convergence domain*

Because $D_m$ is centrally symmetric and convex,

$$\frac{D_m - D_m}{2} = D_m.$$

The support of the difference variables therefore satisfies

$$\max_{(\xi,\eta) \in D_m} (\xi^2 + \eta^2) = \begin{cases} 1, & 1 \leq m \leq 2, \\ 2^{\,1-2/m}, & m > 2. \end{cases} \tag{13}$$

For $1 \leq m \leq 2$, the square-root expansion remains within its convergence disk throughout the integration domain. The rational expansion is directly controlled by $|\beta| < 1$. A common working convergence domain is consequently

$$\boxed{1 \leq m \leq 2, \qquad 0 < \lambda < \sqrt{2}.} \tag{14}$$

At $m = 2$, the square-root singularity occurs at the boundary and is integrable, so the integrated series still converges. For $m > 2$, an open portion of the difference domain satisfies $\xi^2 + \eta^2 > 1$, preventing direct summation of the original square-root series. For $|\beta| > 1$, the rational expansion also fails on part of its support. Boundary cases such as $|\beta| = 1$ require separate examination.

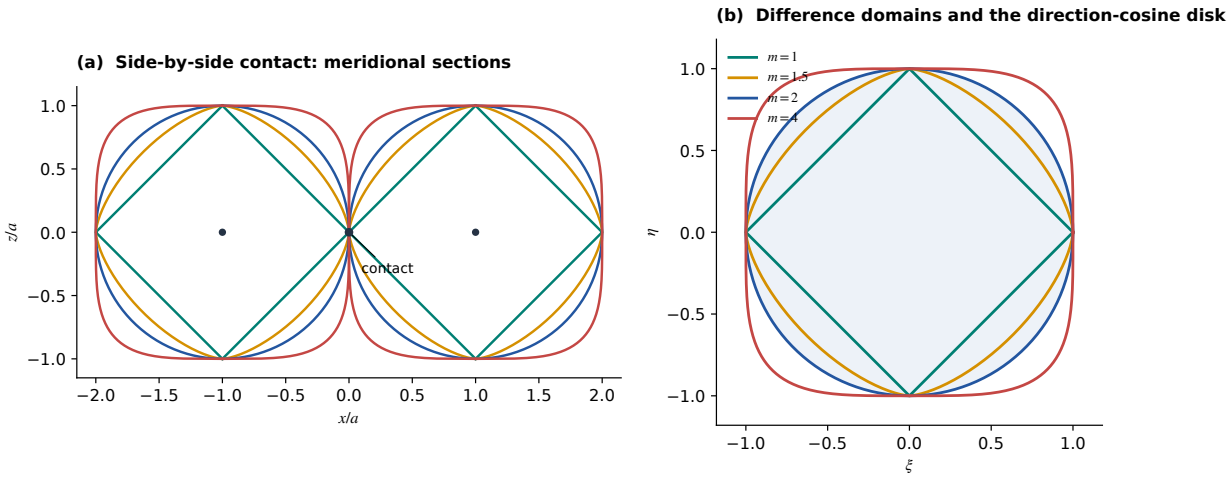



Figure 1: Particle geometry for $\lambda = 1$. (a) Meridional sections of two side-by-side tangent particles. (b) Projected difference domains $D_m$; the shaded unit disk represents admissible direction cosines. The domain remains inside this disk for $1 \le m \le 2$ and extends outside it for $m > 2$.

### *2.5. Direct summation and transcendental integral*

Within Eq. (14),

$$\sum_{p,q,r\ge 0} \frac{(1/2)_{p+q}}{p!\,q!}(r+1)\beta^r \xi^{2p}\eta^{2(q+r)} = \frac{1}{\sqrt{1-\xi^2-\eta^2}}\frac{1}{(1-\beta\eta^2)^2}. \tag{15}$$

Thus

$$F_\Sigma = \frac{\lambda}{16\pi^2 I_m(\lambda)} \int_{D_m}\int_{D_m} \frac{du\,dv\,du'\,dv'}{\sqrt{1-\xi^2-\eta^2}(1-\beta\eta^2)^2}. \tag{16}$$

The infinite sums have been removed. Two integrations can additionally be performed analytically. Define

$$B_m(u) = (1-|u|^m)^{1/m}, \qquad b = \sqrt{1-\frac{(u-u')^2}{4}},$$

$$t_\pm = \frac{B_m(u) \pm B_m(u')}{2}.$$

Then

$$\boxed{F_\Sigma = \frac{\lambda}{2\pi^2 I_m(\lambda)} \int_{-1}^{1}\int_{-1}^{1} [K_{b,\beta}(t_+) - K_{b,\beta}(t_-)]\,du\,du',} \tag{17}$$

where

$$K_{b,\beta}(0) = K'_{b,\beta}(0) = 0, \qquad K''_{b,\beta}(t) = \frac{1}{\sqrt{b^2 - t^2}(1 - \beta t^2)^2}.$$

Setting

$$w = \sqrt{b^2 - t^2}, \qquad k = 1 - \beta b^2 > 0$$

gives the explicit kernel

$$\boxed{K_{b,\beta}(t) = \frac{(1+k)t}{2k^{3/2}} \arctan\left(\frac{\sqrt{k}\, t}{w}\right) + \frac{w - b}{2k} + \frac{\mathcal{A}_{\beta,k}(w) - \mathcal{A}_{\beta,k}(b)}{2k},} \tag{18}$$

with

$$\mathcal{A}_{\beta,k}(w) = \begin{cases} \dfrac{\arctan\left(w\sqrt{\beta/k}\right)}{\sqrt{k\beta}}, & \beta > 0, \\ w, & \beta = 0, \\ \dfrac{\operatorname{artanh}\left(w\sqrt{-\beta/k}\right)}{\sqrt{-k\beta}}, & \beta < 0. \end{cases} \tag{19}$$

For $\lambda = 1$, $\beta = 0$, and

$$\boxed{K_{b,0}(t) = t \arcsin\left(\frac{t}{b}\right) + \sqrt{b^2 - t^2} - b.} \tag{20}$$

Equations (17)–(20) replace the infinite moment series by a two-dimensional integral with an explicit algebraic and transcendental kernel.

*2.6. Multivariate Fox–Wright representation*

Introduce

$$\alpha = \frac{1}{m}, \qquad \delta = \frac{2}{m},$$
$$\mathbf{x} = \left(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{\beta}{4}, \frac{\beta}{4}\right).$$

For $\mathbf{k} = (k_1, \ldots, k_6)$, write

$$P = k_1 + k_2, \qquad Q = k_3 + k_4, \qquad R = k_5 + k_6.$$

For $e = 0, 1$, define

$$\begin{aligned} \mathcal{Q}_e(\mathbf{k}) = {} & \frac{\Gamma(\frac{1}{2} + e + P + Q)\Gamma(2 + e + R)}{\Gamma(1 + P)\Gamma(1 + e + Q)\Gamma(1 + e + R)} \\ & \times \frac{\Gamma(1 + 2P)\Gamma(1 + 2e + 2Q)\Gamma(1 + 2e + 2R)}{\Gamma(1 + 2k_1)\Gamma(1 + 2k_2)\displaystyle\prod_{j=3}^{6} \Gamma(1 + e + 2k_j)} \\ & \times \frac{\Gamma(\alpha + \delta k_1)\Gamma(\alpha + \delta k_2)}{\Gamma(1 + 2\alpha + \delta(e + k_1 + k_3 + k_5))} \\ & \times \frac{\Gamma(\alpha + \delta(e + k_3 + k_5))\,\Gamma(\alpha + \delta(e + k_4 + k_6))}{\Gamma(1 + 2\alpha + \delta(e + k_2 + k_4 + k_6))}. \end{aligned} \tag{21}$$

The corresponding Fox–Wright functions are

$$\Psi_e^{(6)}(\mathbf{x};m)=\sum_{\mathbf{k}\geq 0}\mathcal{Q}_e(\mathbf{k})\prod_{j=1}^{6}\frac{\Gamma(1+k_j)x_j^{k_j}}{k_j!}. \tag{22}$$

The factors $\Gamma(1+k_j)=k_j!$ express the ordinary multiple series in generalized Wright form. Equation (21) specifies all parameters explicitly.

The view-factor series becomes

$$\boxed{F_\Sigma(m,\lambda)=\frac{\lambda}{\pi^{5/2}m^4I_m(\lambda)}\left[\Psi_0^{(6)}(\mathbf{x};m)+\frac{\beta}{16}\Psi_1^{(6)}(\mathbf{x};m)\right].} \tag{23}$$

The prefactor follows from

$$G_m\frac{\Gamma(1+2/m)}{\Gamma^2(1/m)}=\frac{1}{m^2}.$$

The two terms in Eq. (23) arise from parity. The powers of $V-V'$ generated by the square-root and rational factors contribute either even powers separately or odd powers whose product is even. These possibilities correspond to $e=0$ and $e=1$. Both contributions are required for a general aspect ratio. The second vanishes when $\lambda=1$.

*2.7. Fox H continuation*

A special-function series retains the convergence restrictions of its defining sum. Mellin–Barnes representations provide continuation subject to the appropriate contour and branch conditions [16–19]. Define

$$\boxed{H_e^{(6)}(\mathbf{z};m)=\frac{1}{(2\pi i)^6}\int_{\mathcal{L}_1}\cdots\int_{\mathcal{L}_6}\mathcal{Q}_e(-\mathbf{s})\prod_{j=1}^{6}\Gamma(s_j)\Gamma(1-s_j)z_j^{-s_j}\,d\mathbf{s}.} \tag{24}$$

The contours initially separate the relevant pole families in admissible parameter sectors. This explicit Gamma kernel fixes the multivariate Fox $H$ convention.

Taking residues at $s_j=-k_j$ gives

$$H_e^{(6)}(-\mathbf{x};m)=\Psi_e^{(6)}(\mathbf{x};m)$$

in the series convergence domain.

Introduce an auxiliary parameter $\tau$:

$$\mathbf{x}(\tau)=\left(\frac{\tau}{4},\frac{\tau}{4},\frac{\tau}{4},\frac{\tau}{4},\frac{\beta}{4},\frac{\beta}{4}\right).$$

The continued function is

$$\boxed{\mathcal{V}(\tau;m,\lambda)=\frac{\lambda}{\pi^{5/2}m^4I_m(\lambda)}\left[H_0^{(6)}(-\mathbf{x}(\tau);m)+\frac{\tau\beta}{16}H_1^{(6)}(-\mathbf{x}(\tau);m)\right].} \tag{25}$$

For sufficiently small $\tau$,

$$\mathcal{V}(\tau;m,\lambda)=F_0\left\langle\frac{1}{\sqrt{1-\tau(\xi^2+\eta^2)}(1-\beta\eta^2)^2}\right\rangle.$$

Within Eq. (14), the required value is $\mathcal{V}(1;m,\lambda)$. For $m>2$, part of the difference domain crosses the square-root branch cut. The directional geometry in Eq. (8) determines the appropriate boundary value:

$$\boxed{\widehat{F}_{12}(m,\lambda)=\frac{1}{2}\lim_{\epsilon\downarrow 0}\left[\mathcal{V}(1+i\epsilon;m,\lambda)+\mathcal{V}(1-i\epsilon;m,\lambda)\right].} \tag{26}$$

Indeed,

$$\frac{1}{2}\left[(1-r^2+i0)^{-1/2}+(1-r^2-i0)^{-1/2}\right]=\begin{cases}(1-r^2)^{-1/2}, & r<1,\\ 0, & r>1.\end{cases} \tag{27}$$

The boundary-value average therefore recovers the admissible angular support.

The same prescription can be used in Eq. (17). For $|t|>b$, replace the kernel by its linear extension

$$\boxed{K^{\circ}_{b,\beta}(t)=K_{b,\beta}(b)+(|t|-b)\frac{\pi(1+k)}{4k^{3/2}},\qquad k=1-\beta b^2,} \tag{28}$$

and retain Eq. (18) for $|t|\le b$. This is the double antiderivative of the geometrically truncated angular kernel.

When $\lambda=1$, the last two Wright variables vanish and the $e=1$ term disappears. Eliminating those indices before forming the contour integral gives

$$\boxed{\mathcal{V}(\tau;m,1)=\frac{H_0^{(4)}\left(-\frac{\tau}{4},-\frac{\tau}{4},-\frac{\tau}{4},-\frac{\tau}{4};m\right)}{\pi^{5/2}m^4 I_m(1)}.} \tag{29}$$

Thus the equal-axis problem requires four Fox $H$ variables.

## 3. Results and discussion

### *3.1. Analytical benchmark limits*

For two tangent spheres, $m=2$ and $\lambda=1$,

$$\boxed{\widehat{F}_{12}(2,1)=\frac{1}{16}\,{}_3F_2\left(\begin{matrix}\frac{1}{2},1,\frac{3}{2}\\ 2,3\end{matrix};1\right)=\frac{1}{2}-\frac{4}{3\pi}=0.07558681842\ldots.} \tag{30}$$

This recovers the exact spherical view factor [12].

For the equal-axis double-cone endpoint,

$$\boxed{\widehat{F}_{12}(1,1)=\frac{2-\sqrt{2}}{\pi}-\frac{10-4\sqrt{2}}{3\pi^2}=0.03977739156\ldots.} \tag{31}$$

In the cylindrical limit, $m\to\infty$, the fixed reference projection becomes a square, with

$$C_{\infty}(2\xi,2\eta)=4(1-|\xi|)(1-|\eta|)$$

inside the direction-cosine disk. Since $I_{\infty}(1)=3/2$,

$$\boxed{\widehat{F}_{12}(\infty,1)=\frac{8}{9\pi^2}=0.09006327435\ldots.} \tag{32}$$

Equations (31) and (32) are exact evaluations of the projected-overlap model. Their physical approximation errors are examined below.

*3.2. Individual terms and series convergence*

Figure 2 presents the individual terms and accumulated truncation errors for $\lambda = 1$. The terms decrease for $m = 1, 1.5, 2$. For $m = 4$, an initial decrease is followed by rapid growth. Consequently, the first few partial sums can appear plausible even when the infinite series diverges.

For spheres, the coefficient has the explicit form

$$R_{2n}^{(2)} = \frac{(1/2)_n (1)_n (3/2)_n}{(2)_n (3)_n \, n!} \sim \frac{4}{\pi n^3}. \tag{33}$$

Since $F_0 = 1/16$,

$$\boxed{\widehat{F}_{12}(2,1) - F_N(2,1) \sim \frac{1}{8\pi N^2}, \qquad N \to \infty.} \tag{34}$$

The spherical endpoint therefore has algebraic convergence, with a quantitative truncation-error estimate. For $m > 2$, positivity of the equal-axis series and the support of the difference variables give

$$\boxed{\lim_{n\to\infty} \left(R_{2n}^{(m)}\right)^{1/n} = 2^{\,1-2/m} > 1, \qquad \lambda = 1.} \tag{35}$$

The divergence is controlled by particle geometry.

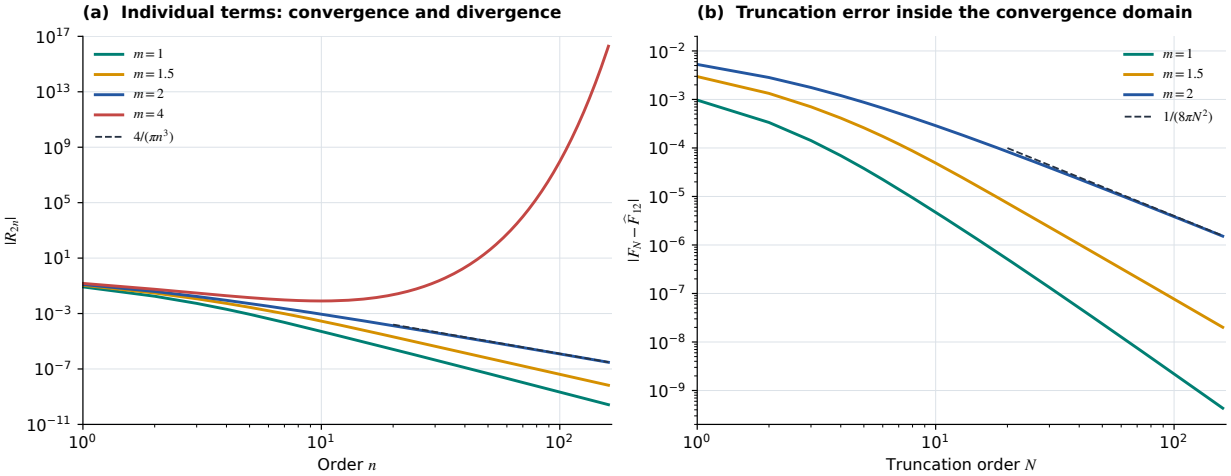



Figure 2: Series behavior for $\lambda = 1$. (a) Individual terms $R_{2n}$, including the divergent $m = 4$ case. (b) Accumulated truncation errors in the convergence domain. Dashed lines show the analytical spherical asymptotes.

Table 1: Partial sums and directly summed results for $\lambda = 1$.

| $m$ | $F_{20}$ | $F_{50}$ | $F_{100}$ | Summed or continued model |
|---|---|---|---|---|
| 1 | 0.03977688 | 0.03977737 | 0.0397773894 | 0.0397773916 |
| 1.5 | 0.06306634 | 0.06307309 | 0.0630735643 | 0.0630736411 |
| 2 | 0.07550271 | 0.07557196 | 0.0755829752 | 0.0755868184 |
| 4 | 0.10240155 | 9.35072695 | $2.47349 \times 10^7$ | 0.09188442 |

For $m = 1.5$, the absolute error at $N = 100$ is approximately $7.7 \times 10^{-8}$. For spheres, it remains approximately $3.8 \times 10^{-6}$, consistent with Eq. (34). Increasing the truncation order for $m = 4$ produces exponential growth rather than improved accuracy.

*3.3. Verification of direct summation and the Fox–Wright form*

The direct summation was checked against the original moment hierarchy for several aspect ratios within Eq. (14). Figure 3(a) compares the summed integral with $N = 100$ partial sums for $m = 1.5$ and $m = 2$. Their agreement follows the truncation errors identified in Fig. 2.

The Gamma reindexing was also checked independently. For $m = 1, 1.5, 2, 4$, all original coefficients with $p + q + r \leq 6$ were compared with Eq. (21), giving a maximum relative mismatch of $6.7 \times 10^{-15}$.

Complete finite Fox–Wright sums were evaluated separately through order $N = 12$. For example, at $m = 1.5, \lambda = 0.8$,

$$F_{12}^{\text{series}} = 0.0509357676948864,$$

$$F_{12}^{\text{Wright}} = 0.0509357676948864.$$

Across the four tested finite-sum cases, the relative mismatch remained below $8 \times 10^{-16}$.

The parity contribution in Eq. (23) is numerically relevant. Omitting the $e = 1$ term in the same $m = 1.5, \lambda = 0.8, N = 12$ calculation gives 0.0504827541, demonstrating that the odd-times-odd contribution cannot generally be discarded.

These checks verify the algebraic equality of the moment and Gamma-function representations.

*3.4. Verification of Fox H continuation*

Continuation was tested with a reduced Barnes integral obtained after integrating the geometric sector through its weighted Mellin moment. Define

$$\mathcal{M}_{m,\beta}(s) = \iint_{D_m} \frac{C_m(2\xi, 2\eta)(\xi^2 + \eta^2)^s}{(1 - \beta\eta^2)^2}\, d\xi\, d\eta. \tag{36}$$

The boundary-value average in Eq. (26) can then be evaluated as

$$\boxed{\widehat{F}_{12} = \frac{\lambda}{4\pi^{5/2} I_m(\lambda)} \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \Gamma(-s)\Gamma\left(\frac{1}{2} + s\right) \cos(\pi s)\mathcal{M}_{m,\beta}(s)\, ds,} \tag{37}$$

where $-1/2 < c < 0$.

Equation (37) evaluates the continued Barnes kernel independently of the real inverse-trigonometric expression. It also avoids direct six-dimensional contour quadrature. The calculations used $c = -1/4$, with the contour cutoff increased from $T = 16$ to 512.

Table 2: Independent continuation checks at $T = 512$.

| $m$ | $\lambda$ | Summed/geometric-boundary result | Reduced Barnes result | Absolute difference |
|---|---|---|---|---|
| 2 | 1 | 0.0755868185 | 0.0755869593 | $1.41 \times 10^{-7}$ |
| 2 | 2 | 0.1108058337 | 0.1108059365 | $1.03 \times 10^{-7}$ |
| 4 | 1 | 0.0918844236 | 0.0918844692 | $4.56 \times 10^{-8}$ |
| 4 | 2 | 0.1244940767 | 0.1244941463 | $6.97 \times 10^{-8}$ |
| $\infty$ | 1 | 0.0900632743 | 0.0900632863 | $1.19 \times 10^{-8}$ |

The cases $m = 4$ and $\lambda = 2$ test continuation beyond the original shape and aspect-ratio restrictions, respectively. Both give stable finite results.

An additional spherical check used the exact radial Mellin moment. Combining contour cutoffs $T = 512$ and 1024 through the known $T^{-2}$ tail gave

$$F_{\text{Barnes}} = 0.0755868184220849,$$

with an absolute difference of approximately $4.7 \times 10^{-13}$ from Eq. (30).

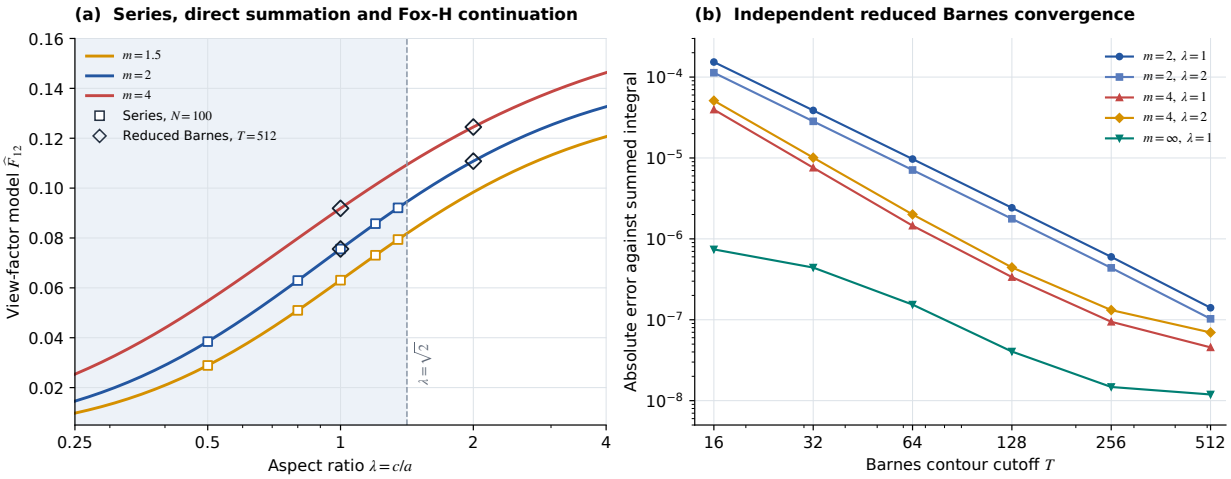



Figure 3: Verification of the three representations. (a) Lines show the directly summed expression, with the geometric boundary value applied when required; squares show $N = 100$ moment sums; diamonds show independent reduced Barnes calculations. Shading marks $0 < \lambda < \sqrt{2}$, the aspect-ratio portion of the working convergence domain for $m \leq 2$. The $m = 4$ curve requires continuation throughout. (b) Convergence of the reduced Barnes calculation with contour cutoff.

The numerical results therefore verify three separate steps: the moment coefficients, the directly summed representation, and the continued Barnes boundary value.

### *3.5. Diffuse-ray comparison and geometric approximation error*

Independent diffuse-ray calculations were performed on the actual particle surfaces. Emission directions followed the cosine distribution, and ray intersections used the original particle equation.

Each estimate used four independently scrambled Sobol sequences. The three initial benchmarks retained $2^{21}$ rays per scramble; the additional cases used $2^{20}$. Reported standard errors were estimated across the four scrambles.

Define the geometric-model discrepancy as

$$\varepsilon_{\text{geom}} = \left( \frac{\widehat{F}_{12}}{F_{\text{ray}}} - 1 \right) \times 100\%. \tag{38}$$

The spheroidal cases agree with the independent estimates within sampling uncertainty, consistent with the exactness of the affine projection formulation at $m = 2$.

For the sampled equal-axis shapes with $1 \leq m \leq 4$, the absolute relative discrepancy remains below approximately 4.5%. The error changes sign across the spherical geometry: the model slightly overpredicts the sharper particles with $m < 2$ and underpredicts the blockier particles with $m > 2$.

The discrepancy grows toward the cylindrical limit, reaching approximately 20.2%. The model also develops a weak maximum near $m \approx 7$–$8$. This maximum should not be interpreted as an optimum of the exact physical view factor: it is sensitive to the fixed-silhouette approximation, and the ray results continue to increase over the corresponding sampled shapes.

Fox $H$ continuation resolves the failure of the original series and preserves the stated projected-overlap model. The remaining physical error belongs to the geometric closure. These two issues are separately quantified by Tables 2 and 3.

Table 3: Analytical model and independent physical view-factor estimates.

| $m$ | $\lambda$ | Analytical model | Diffuse-ray estimate ± SE | $\varepsilon_{\mathrm{geom}}$ |
|---|---|---|---|---|
| 2 | 0.5 | 0.03845966 | $0.038450 \pm 0.000009$ | +0.03% |
| 1 | 1 | 0.03977739 | $0.038477 \pm 0.000008$ | +3.38% |
| 1.25 | 1 | 0.05338940 | $0.051549 \pm 0.000032$ | +3.57% |
| 1.5 | 1 | 0.06307364 | $0.061755 \pm 0.000029$ | +2.14% |
| 1.75 | 1 | 0.07015702 | $0.069587 \pm 0.000014$ | +0.82% |
| 2 | 1 | 0.07558682 | $0.075578 \pm 0.000008$ | +0.01% |
| 2.5 | 1 | 0.08333357 | $0.084025 \pm 0.000018$ | −0.82% |
| 3 | 1 | 0.08774262 | $0.089533 \pm 0.000006$ | −2.00% |
| 4 | 1 | 0.09188442 | $0.096184 \pm 0.000018$ | −4.47% |
| 6 | 1 | 0.09405576 | $0.102439 \pm 0.000014$ | −8.18% |
| 8 | 1 | 0.09423055 | $0.105350 \pm 0.000015$ | −10.55% |
| $\infty$ | 1 | 0.09006327 | $0.112818 \pm 0.000006$ | −20.17% |
| 2 | 2 | 0.11080583 | $0.110802 \pm 0.000021$ | +0.003% |

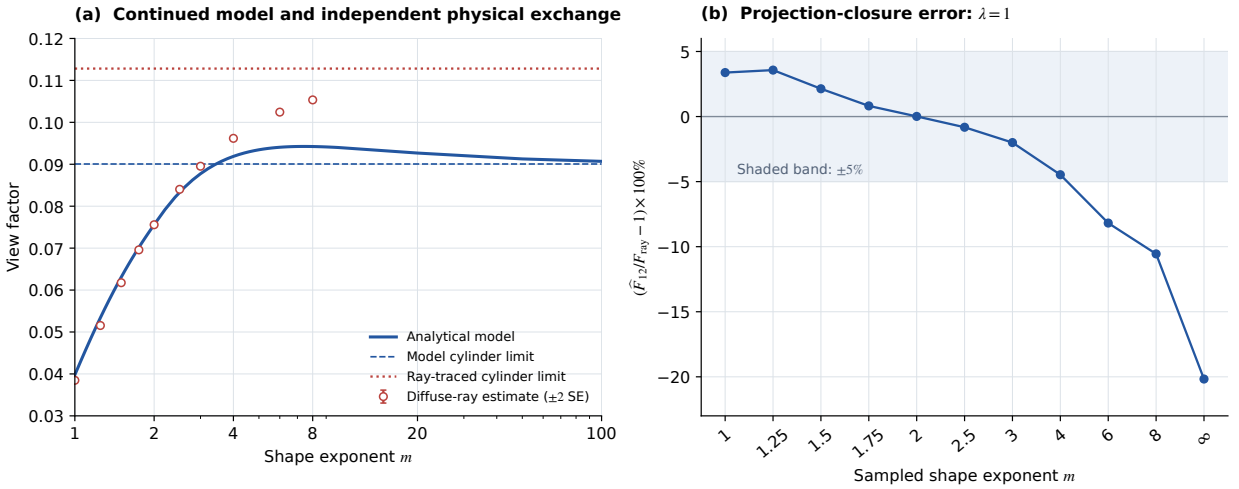



Figure 4: Continued model and independent diffuse-ray results for $\lambda = 1$. (a) Shape dependence and cylindrical limits. (b) Relative projection-model error at the sampled exponents. Error bars indicate twice the estimated standard error; the shaded band denotes ±5%.

## 4. Conclusions

A geometric-moment formulation has been established for radiative exchange between side-by-side tangent axisymmetric superquadric particles. Its main features are:

1. **An explicit connection between particle geometry and special functions.** Superquadric projection moments are expressed through Gamma functions, generating a hierarchy from geometric series to a summed transcendental integral and multivariate Fox $H$ functions.
2. **A geometric explanation of convergence.** The working domain $1 \leq m \leq 2,\ 0 < \lambda < \sqrt{2}$ follows from the projected difference domain and the radiation kernel. For tangent spheres, individual terms decay as $n^{-3}$, giving an explicit $N^{-2}$ truncation-error law.
3. **A branch prescription determined by radiation geometry.** Conjugate Fox $H$ boundary values recover the admissible direction-cosine disk. The continuation therefore has a specified geometric meaning and remains evaluable when the original moment series diverges.
4. **Separate mathematical and physical validation.** Gamma coefficients, finite Fox–Wright sums, directly summed integrals and continued Barnes calculations agree at their respective numerical accuracies. Independent ray tracing recovers exact spheroidal exchange and bounds the discrepancy below approximately 4.5% for the sampled equal-axis shapes with $1 \leq m \leq 4$. Larger deviations toward the cylindrical limit identify where directional silhouette corrections become necessary.

The resulting contribution is a unified analytical structure in which geometry determines the moments, the convergence boundary and the real continuation of the radiation kernel.

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